What is a Data Analysis?

Turning raw facts (numbers, text, dates) **into patterns and answers that help you decide something.**

Where does the data live?

Data usually sits in a table — rows and columns, like a spreadsheet.

A **row** = one record, one event, one entry (e.g. one sale, one customer, one transaction).

A **column** = one type of information about that record (e.g. Country, Product, Revenue).

The two kinds of columns you'll always see

Dimension — a word or label. Describes or groups something (Country, Product, Date, Category). You use it to answer "by what?"

Metric — a number. Something you can count, sum, or average (Revenue, Units Sold, Profit). You use it to answer "how much / how many?"

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The workflow

Collect → Clean → Analyze → Visualize → Decide

Where data analysis is used (industries)

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Data analysis isn't tied to one field — almost every industry runs on it. A few examples:

- **Healthcare** — tracking patient outcomes, hospital wait times, disease spread, drug effectiveness
- **Retail & e-commerce** — sales trends, best-selling products, customer buying patterns (like AfriMart)
- **Finance & banking** — fraud detection, credit risk, investment performance, spending patterns
- **Marketing** — campaign performance, customer segmentation, ad spend vs. return
- **Sports** — player performance stats, team strategy, injury prediction
- **Government & public policy** — census data, crime rates, economic indicators
- **Manufacturing** — production efficiency, defect rates, supply chain tracking
- **Transportation & logistics** — delivery times, route optimization, fuel usage
- **Education** — student performance, dropout risk, course effectiveness
- **Tech & social media** — user engagement, app usage, recommendation systems

One sentence to remember this

Any place that generates numbers and needs better decisions — that's a place for data analysis.

What is a Dashboard?

Turning raw facts (numbers, text, dates) into patterns and answers that help you decide something.

- ❑ Used by managers and decision-makers
- ❑ Focuses on insights, not raw data

Example:
Revenue, Profit, and Sales trends on one screen

546 KB

AfriMart KollyBright Dashboard

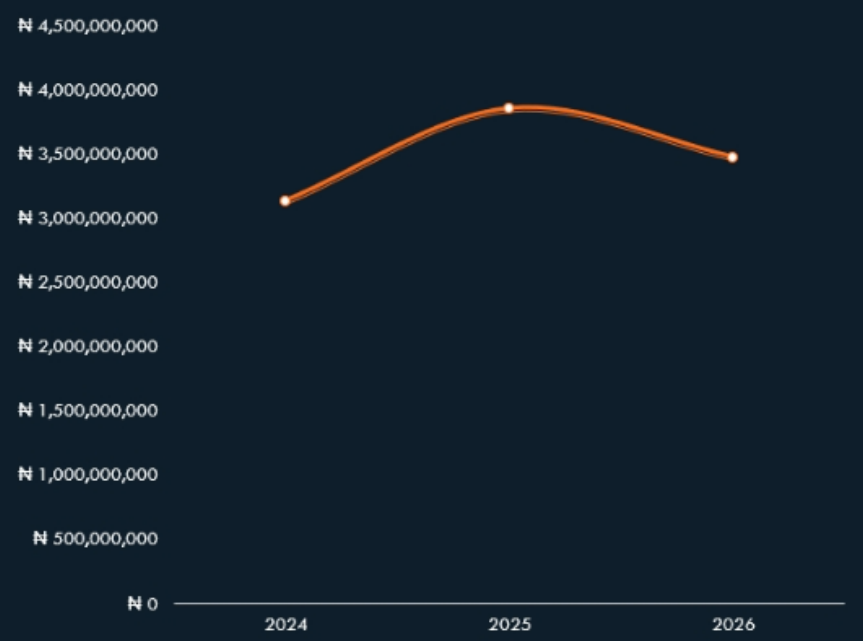
Côte d'Ivoire
Egypt
Ghana
Kenya
Nigeria
Rwanda
Senegal
South Africa
Tanzania
Uganda

2024
2025
2026
<1/2/2024
>12/30/2026

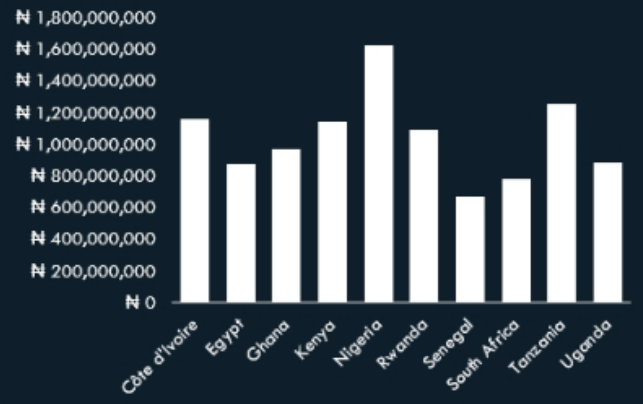
Revenue
₦ 10,447,536,600

Profit
₦ 2,099,369,250

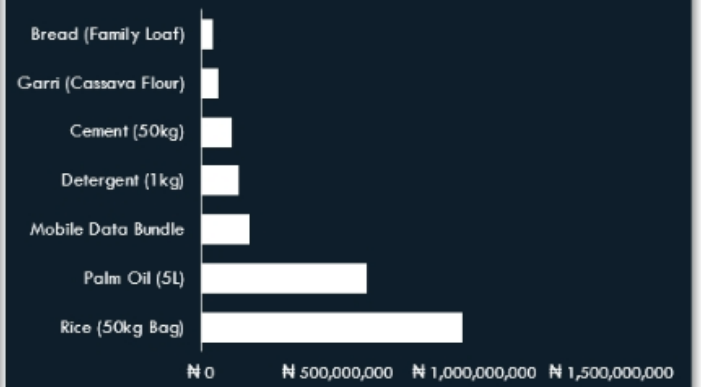
Unit Sold
1017885



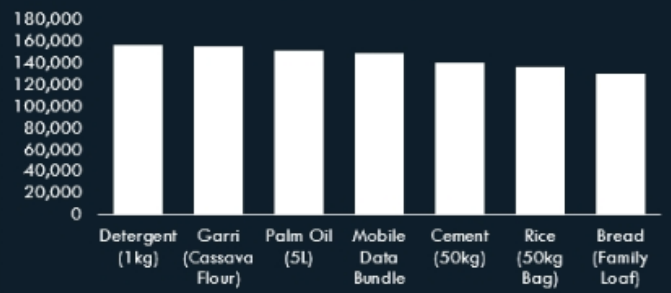
Revenue by Country

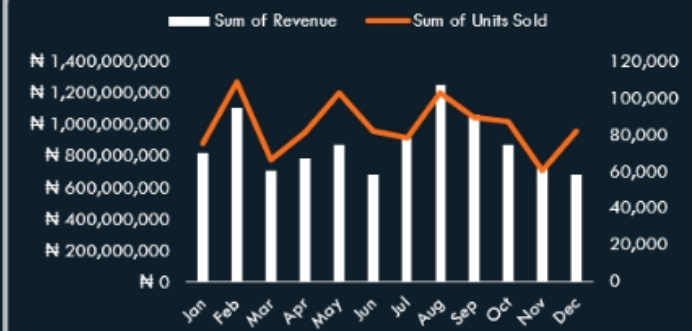


Profit by Product

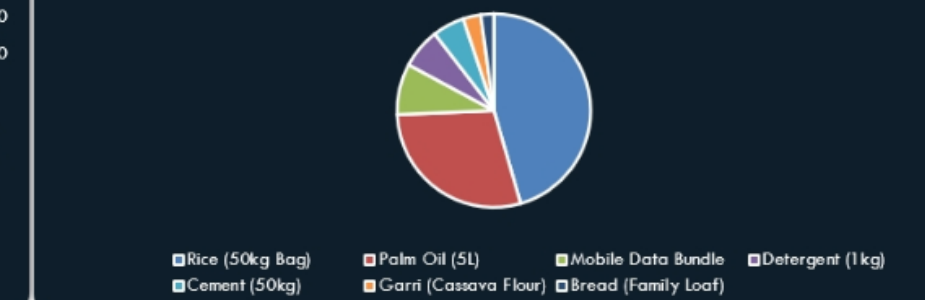


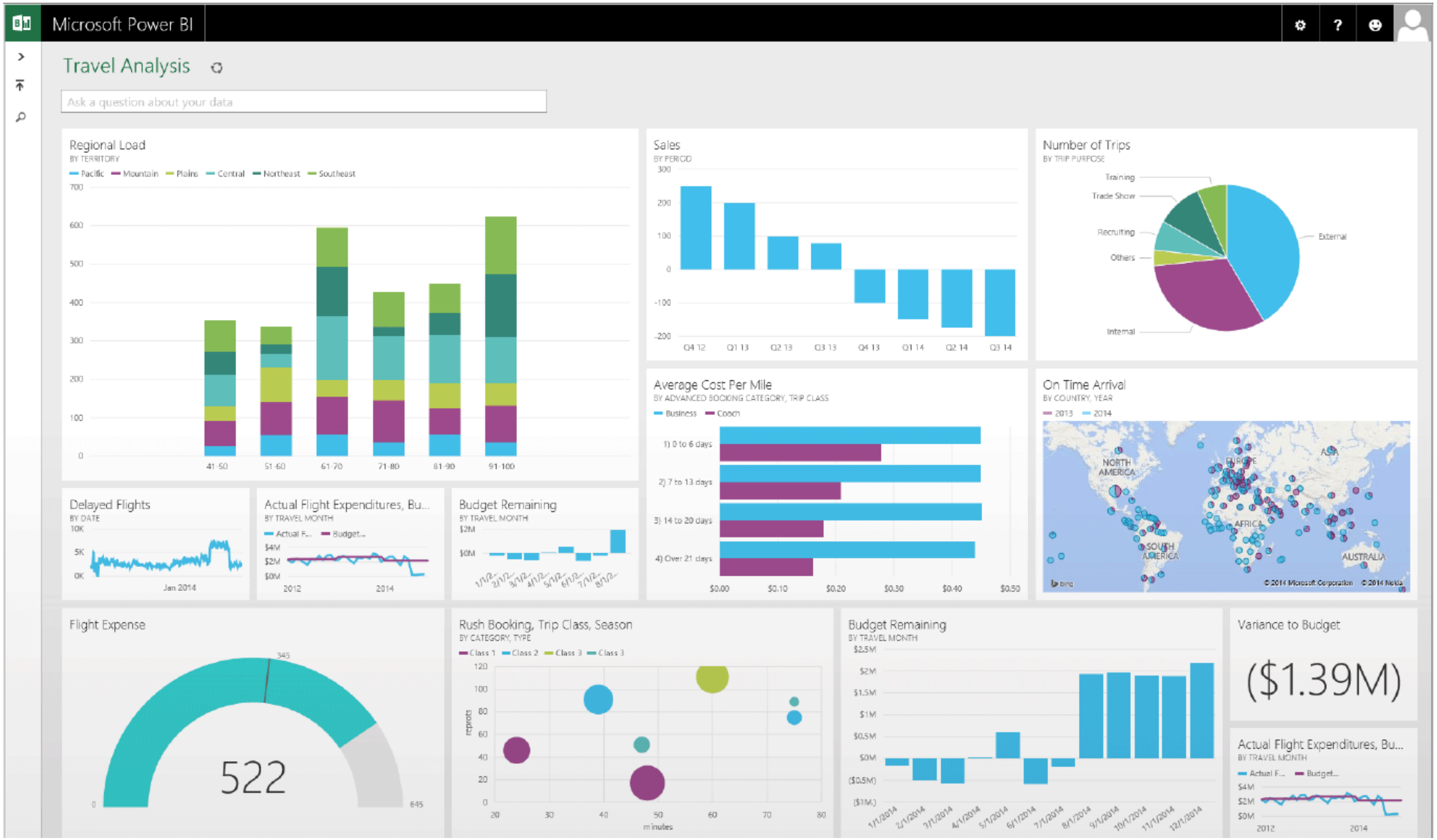
Unit Sold by Product





Top 5 Products

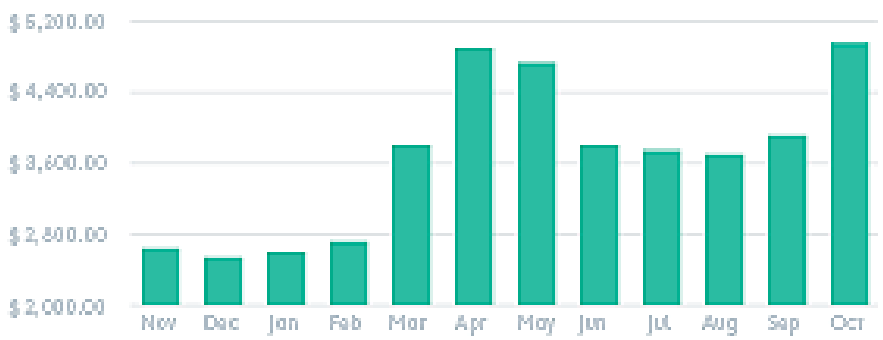




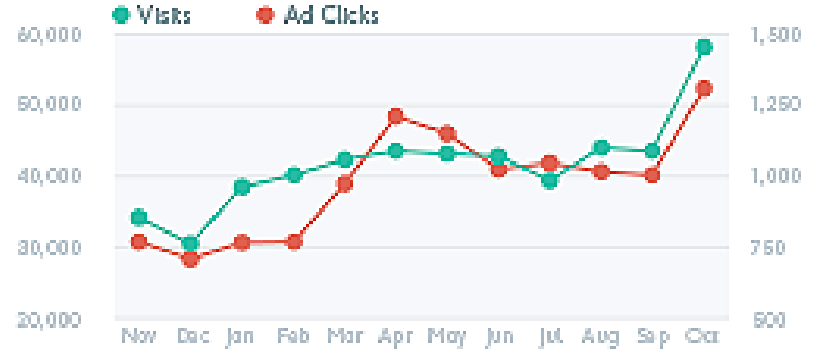
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WIDGETS

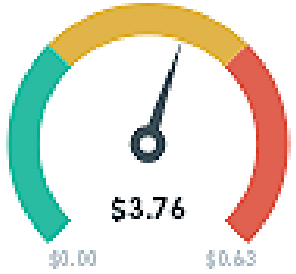
AD COST -TREND WIDGET



VISITS - TREND WIDGET 2 METRICS

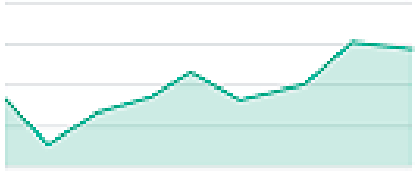


COST PER CLICK (CPC) - GAUGE WIDGET



AD IMPRESSIONS - HISTORIC WIDGET

35,567

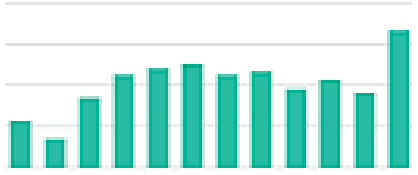


Previous period 0 %

Previous year 3 %

TOTAL PAGEVIEWS

252,292

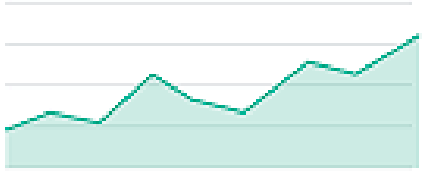


Previous period 34 %

Previous year 62 %

TOTAL VISITS

57,841



Previous period 34 %

Previous year 62 %

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What is a Dashboard?

A dashboard is a single page that shows important business information at a glance.

- ❑ Used by managers and decision-makers
- ❑ Focuses on insights, not raw data

Example:

Revenue, Profit, and Sales trends on one screen

12 dashboard design tips

1. **Be clear about what you're trying to achieve –** your board's purpose will inform its design
2. **Only include what's important –** everything should support your board's intent
3. **Consider data ink ratio –** avoid decorative elements that don't communicate data
4. **Round your numbers –** being overly precise can get in the way of important changes
5. **Use the most efficient visualization –** a good visualization should be understood quickly
6. **Group your related metrics –** make your metrics easy to find
7. **Be consistent –** using the same visualizations and layouts makes comparing easier
8. **Use size and position to show hierarchy –** make it clear to the viewer what's most important
9. **Give your numbers context –** help your viewers know if a number is good, bad, normal or unusual
10. **Use clear labels for your audience –** keep them short and self explanatory
11. **Remember it's for people –** it's ok to break the rules if it increases engagement
12. **Keep evolving your dashboards –** check that your dashboard is encouraging the right behaviour

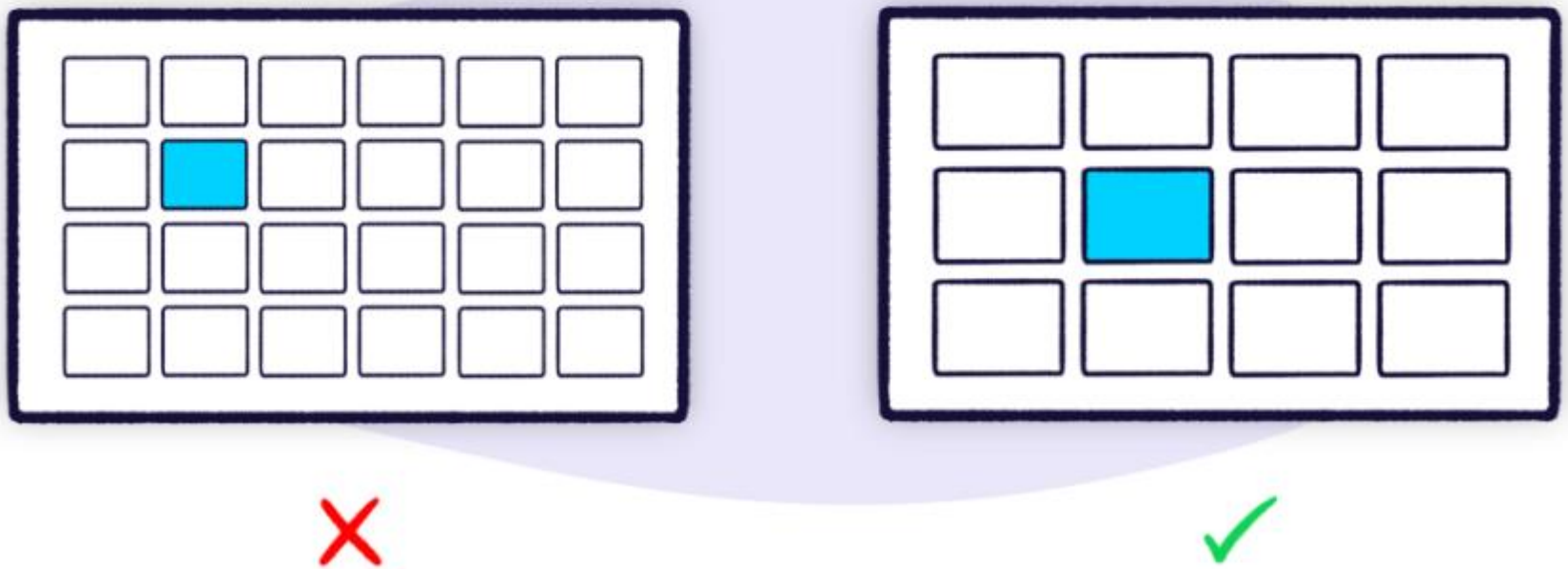
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Be clear about what you're trying to achieve



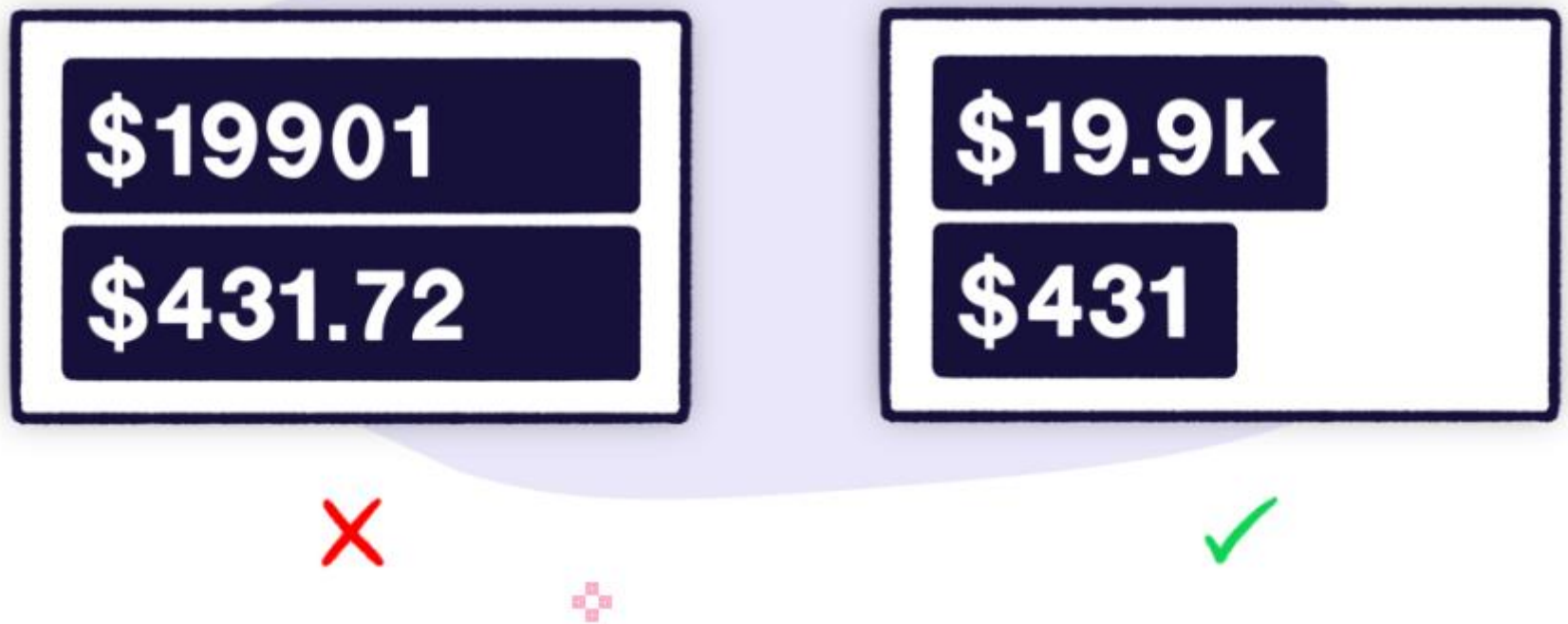
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Include only the most important content



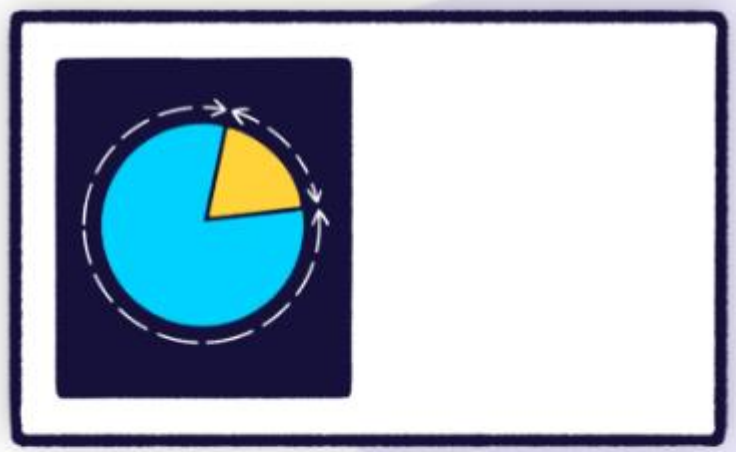
Project Builders
Build Real Experience

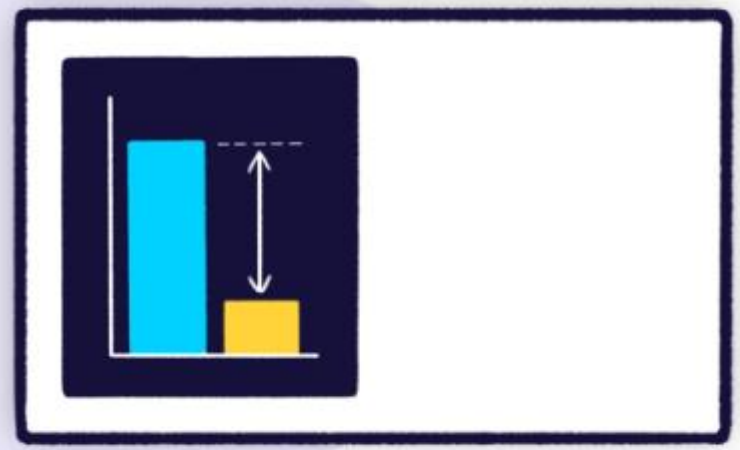
Round your numbers



Project Builders
Build Real Experience

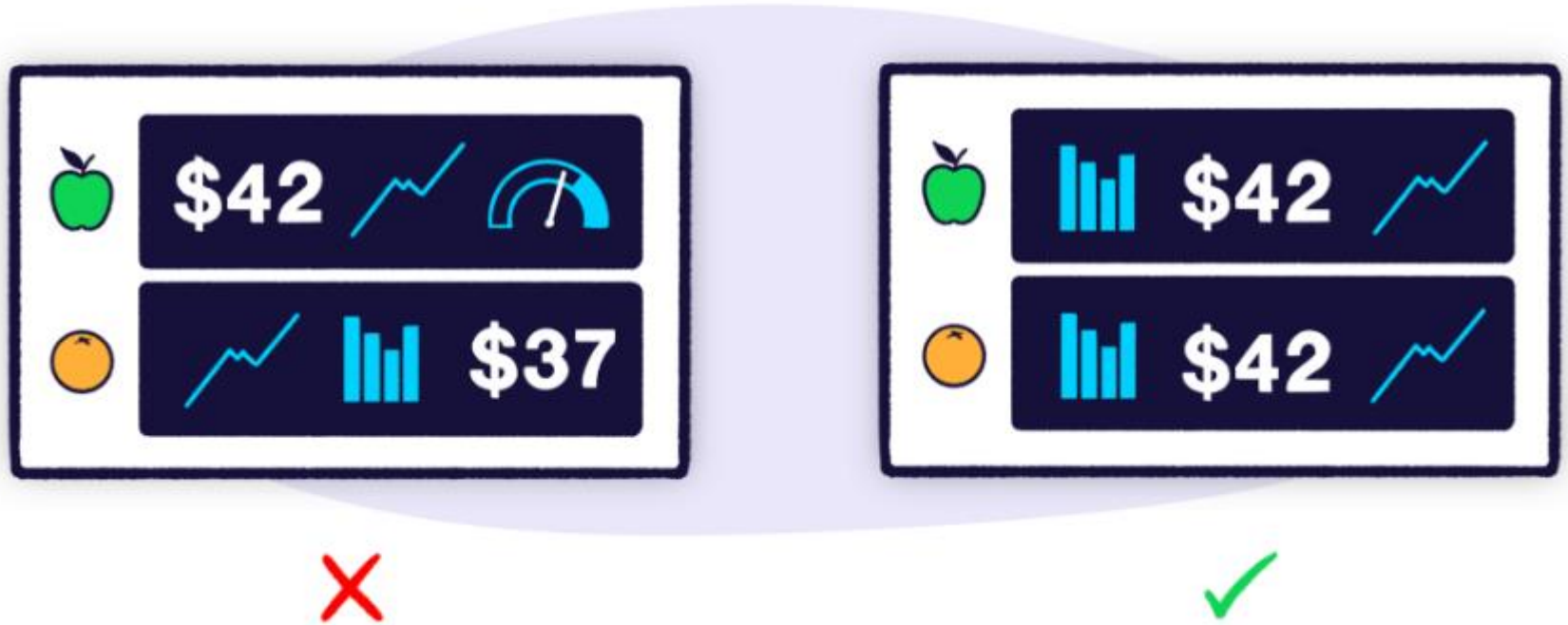
Use the most efficient visualization





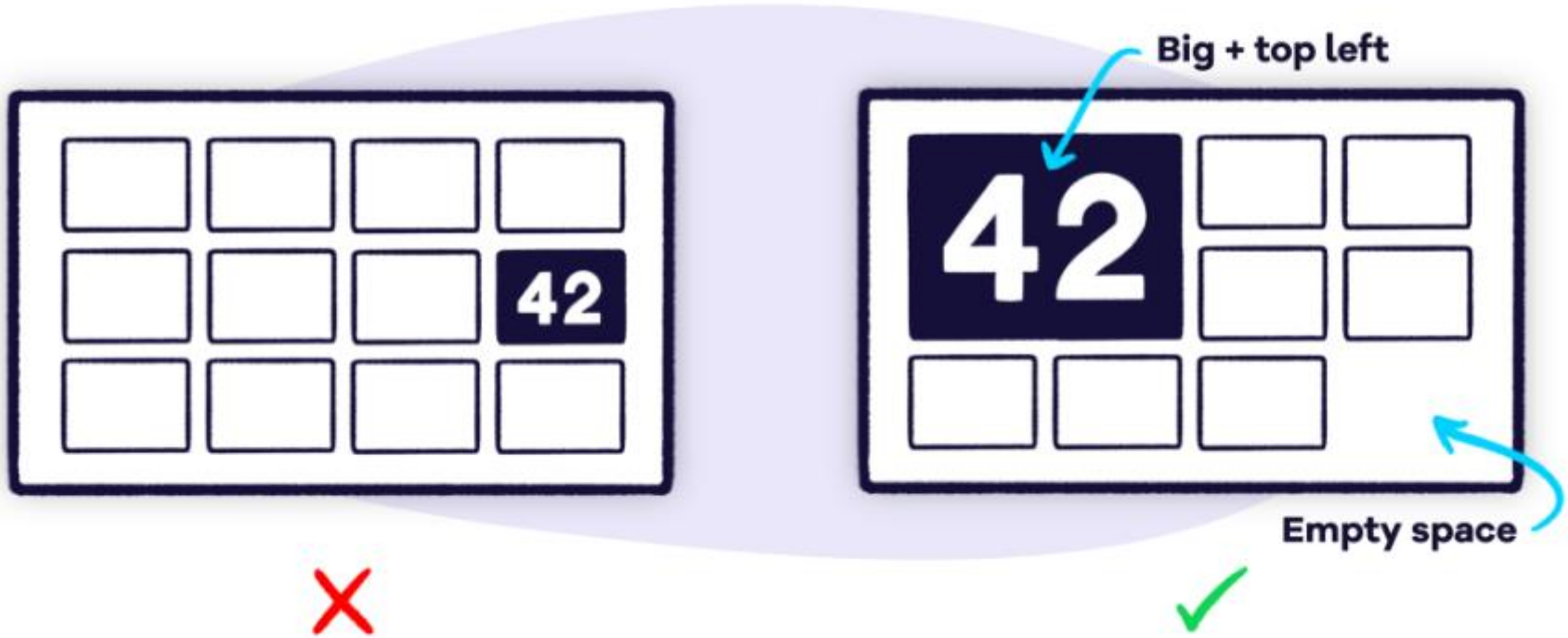
Project Builders
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Be consistent



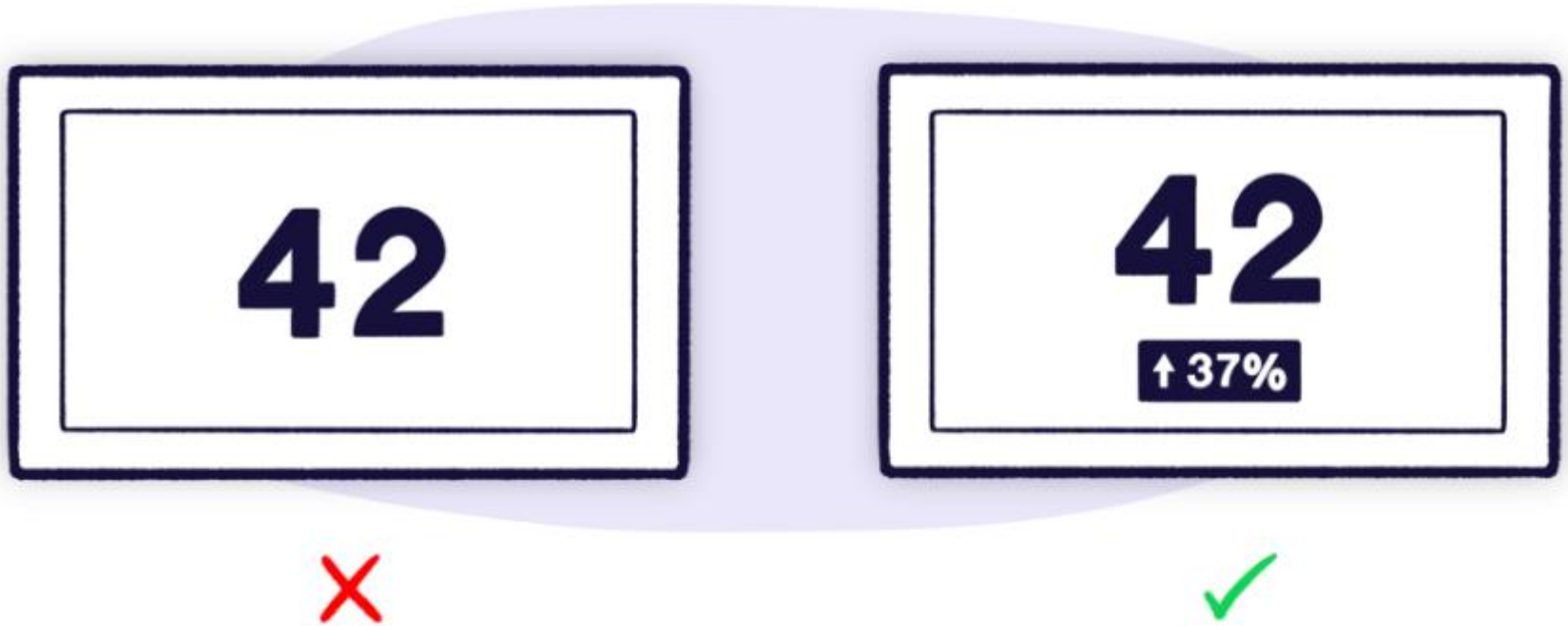
Project Builders
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Use size and position to show hierarchy



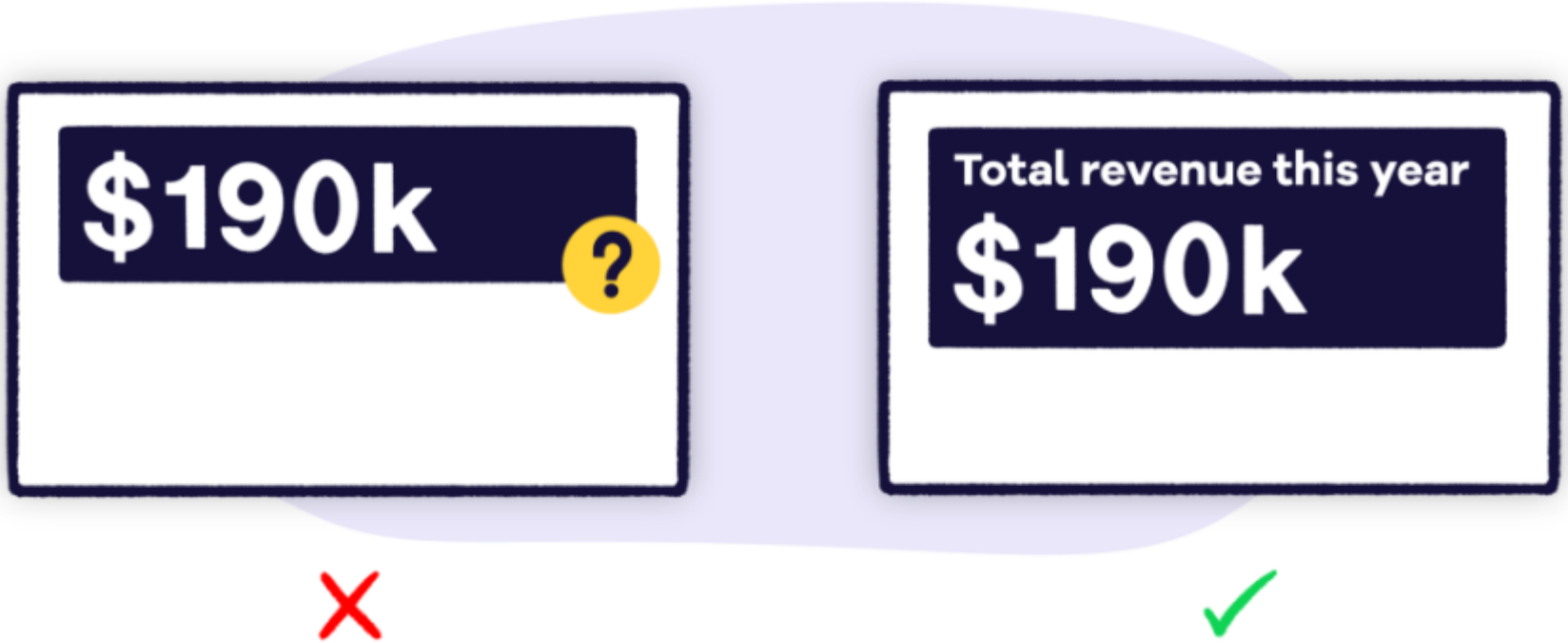
Project Builders
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Give your numbers context



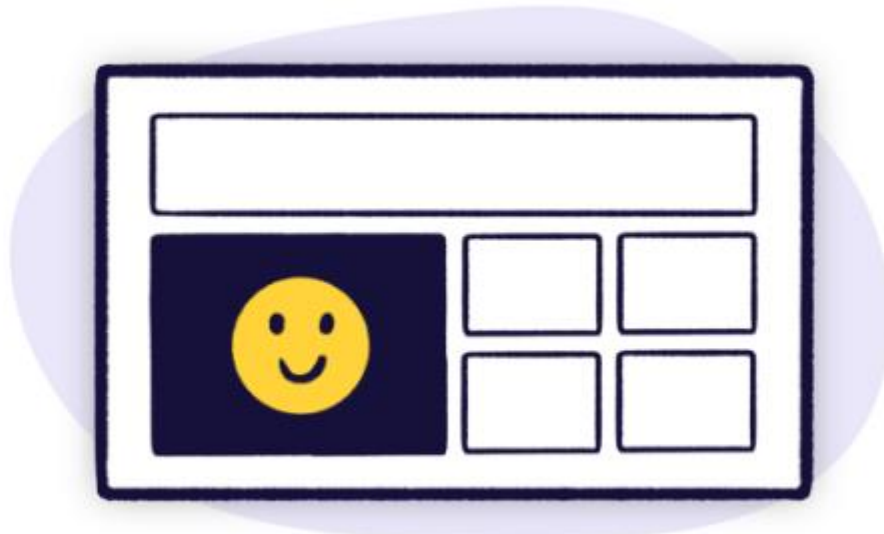
Project Builders
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Use clear labels your audience will understand



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Remember it's for people



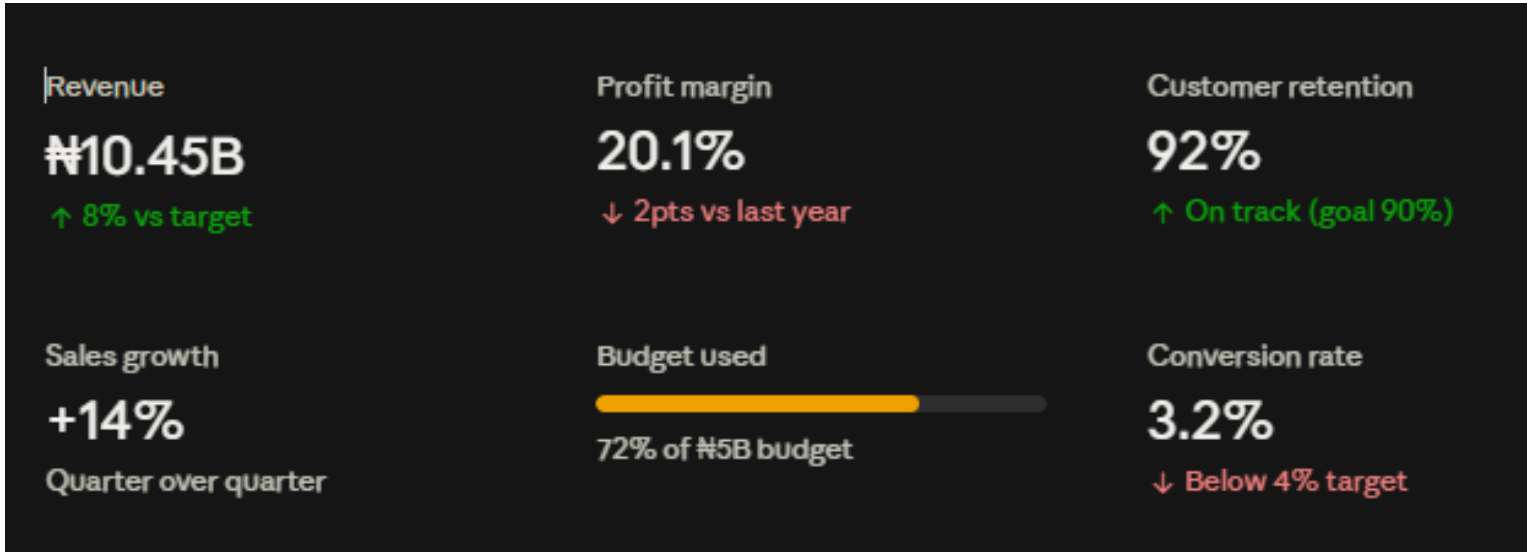
What is a KPI?

KPI (Key Performance Indicator) is a number that tells us **how well the business is doing.**

"KPIs answer the question: *Are we doing well or not?*"

Examples

- ✓ **Total Revenue**
- ✓ **Total Profit**
- ✓ **Total Units Sold**
- ✓ **Profit Margin (%)**



IMPORTANT NOTE

"KPIs are usually **big numbers at the top** of a dashboard."

AfriMart KollyBright Dashboard

AfriMart

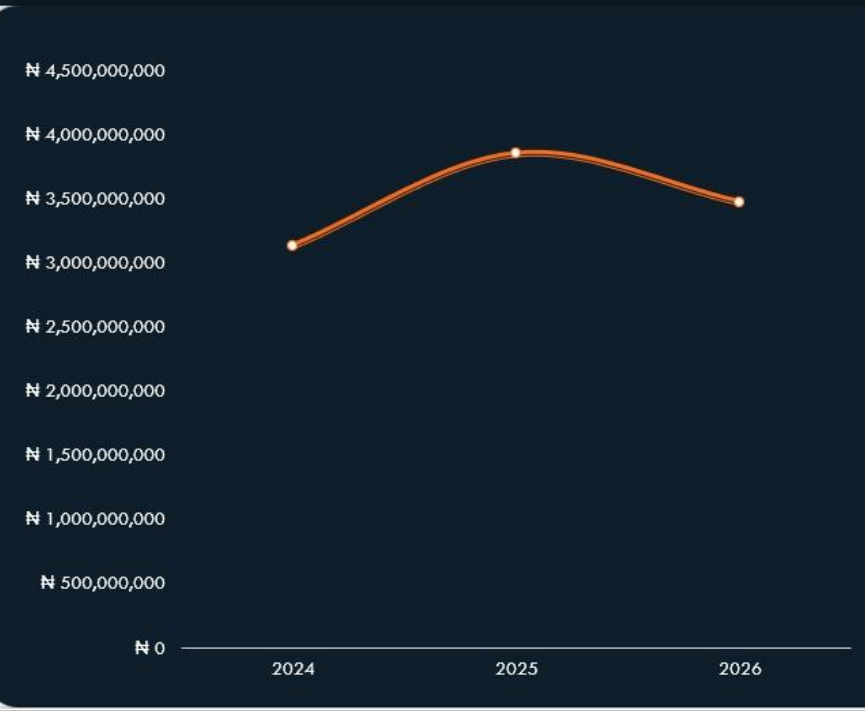
Côte d'Ivoire
Egypt
Ghana
Kenya
Nigeria
Rwanda
Senegal
South Africa
Tanzania
Uganda

2024
2025
2026
<1/2/2024
>12/30/2026

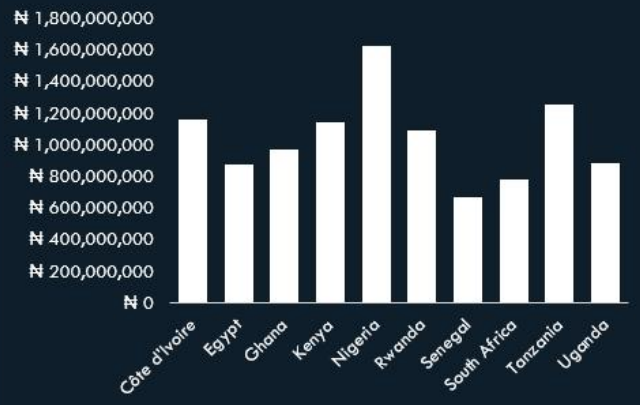
Revenue
₦ 10,447,536,600

Profit
₦ 2,099,369,250

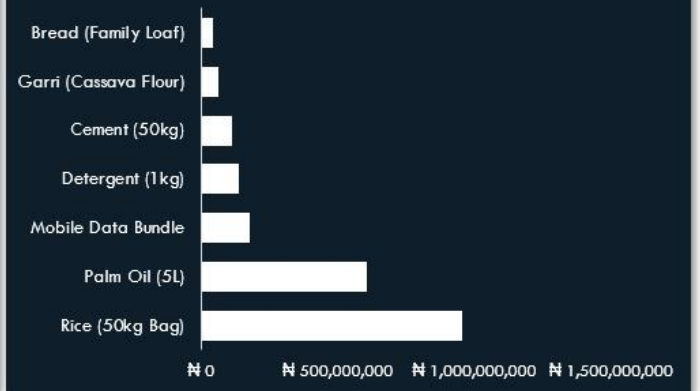
Unit Sold
1017885



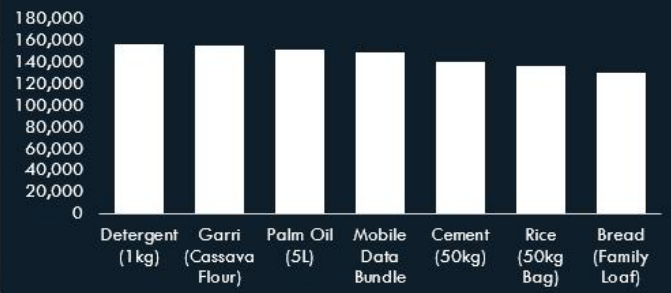
Revenue by Country

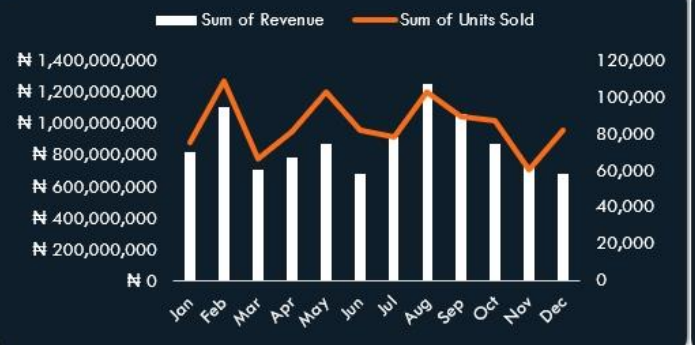


Profit by Product

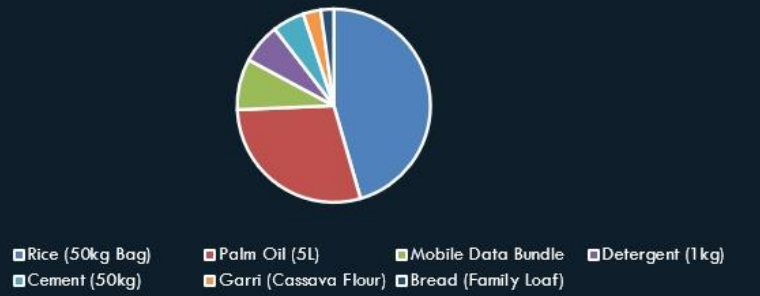


Unit Sold by Product



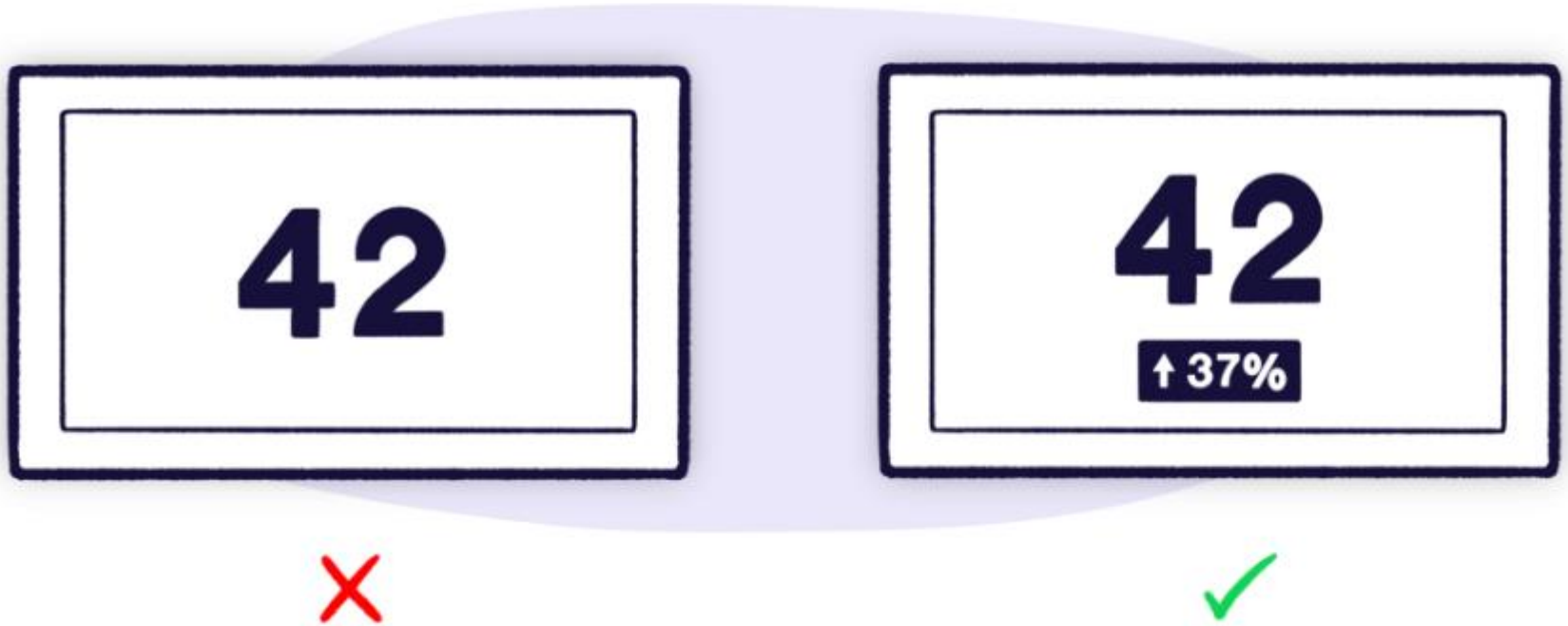


Top 5 Products



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Give your numbers context



Metrics vs Dimensions

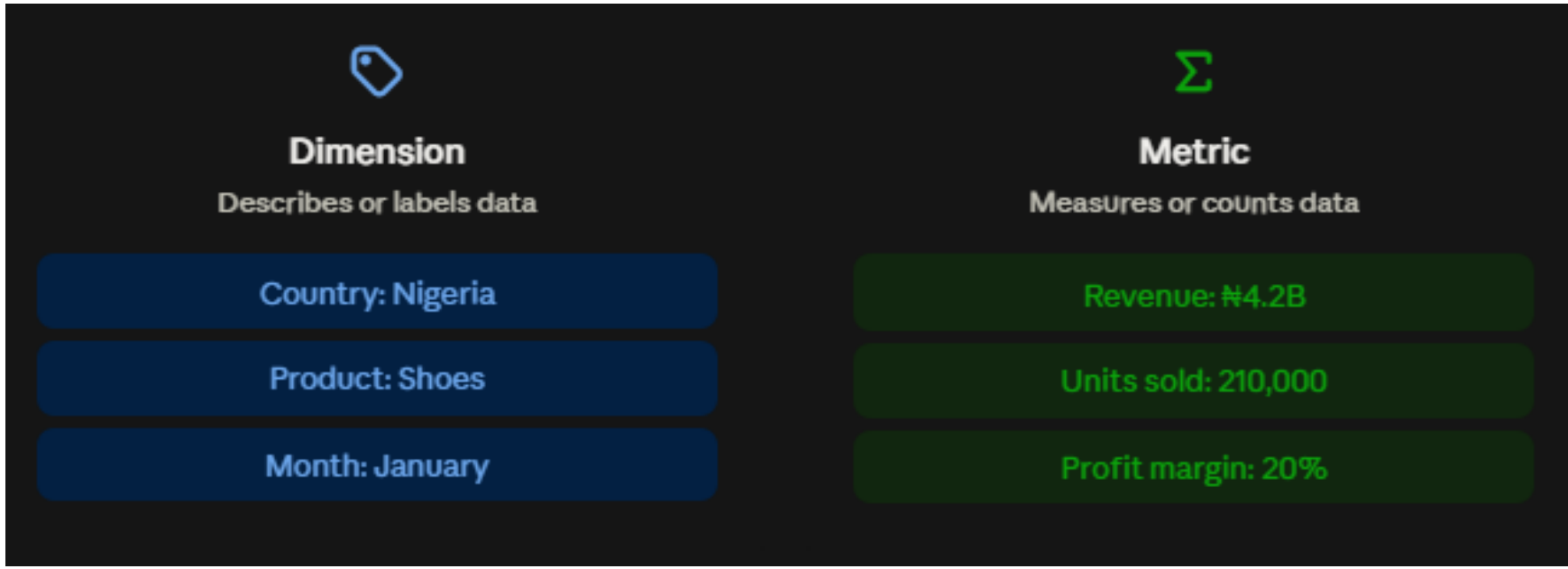
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Metric

*A **number** we calculate.*

Examples:

- Revenue
- Profit
- Units Sold



Dimension

*How we **group or break** down a metric.*

Examples:

- Country
- Product
- Date

Metrics + Dimensions

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Example:

Metric: Revenue

Dimension: Country

➞ Revenue by Country

##################################

Metric: Profit

Dimension: Product

➞ Profit by Product

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How to Name Charts Properly

Rule:

Metric by Dimension

Examples:

- ✓ **Sales** by Country
- ✓ **Revenue** by Product
- ✓ **Profit** by Year
- ✓ **Units Sold** by Month

When There Are Two Dimensions

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If there are two categories, mention both.

Rule:

Metric** by **Dimension and Dimension

Examples:

- ✓ ***Revenue*** by ***Country and Product***
- ✓ ***Profit*** by ***Product and Year***
- ✓ ***Sales*** by ***Country and Year***

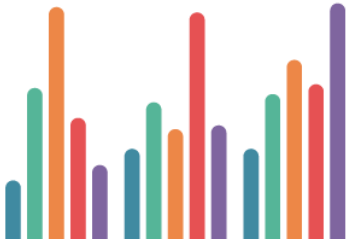
Why We Use Charts

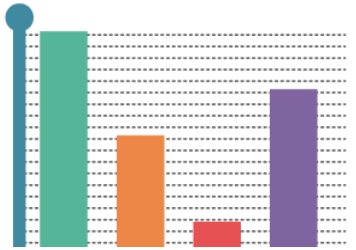
Charts help us understand data quickly.

- ❖ Compare values
- ❖ Spot trends
- ❖ Identify top and bottom performers







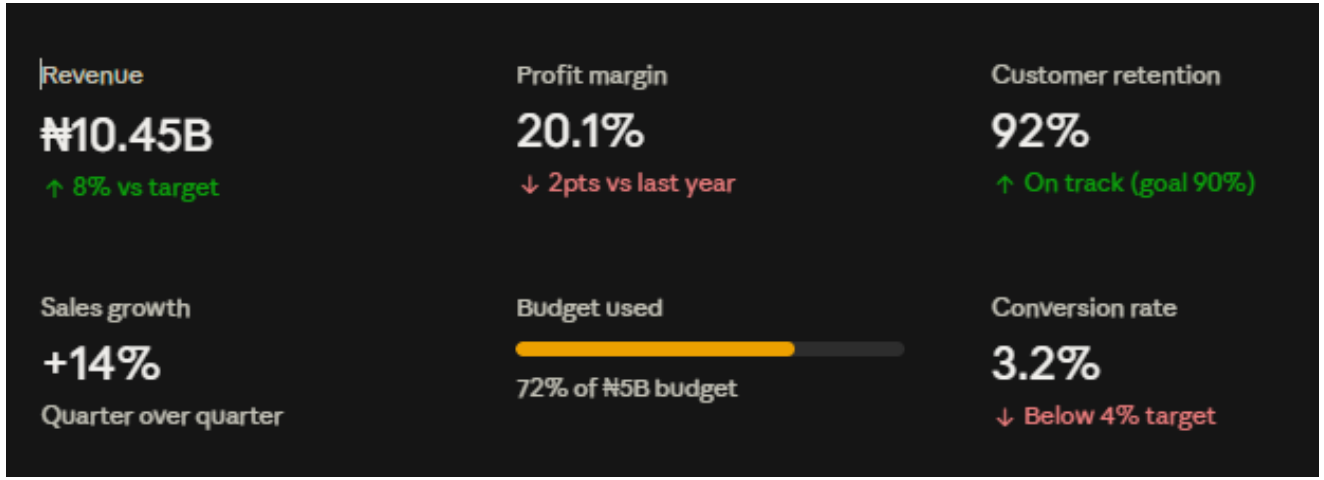


Examples:

- **Bar chart** → Compare products
- **Column chart** → Compare countries
- **Line chart** → Show trends over time

Choose charts based on data type - **use bar charts for comparisons, line charts for trends, and tables for detailed data. Avoid pie charts unless showing parts of a whole.**

KPIs vs Charts



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KPIs

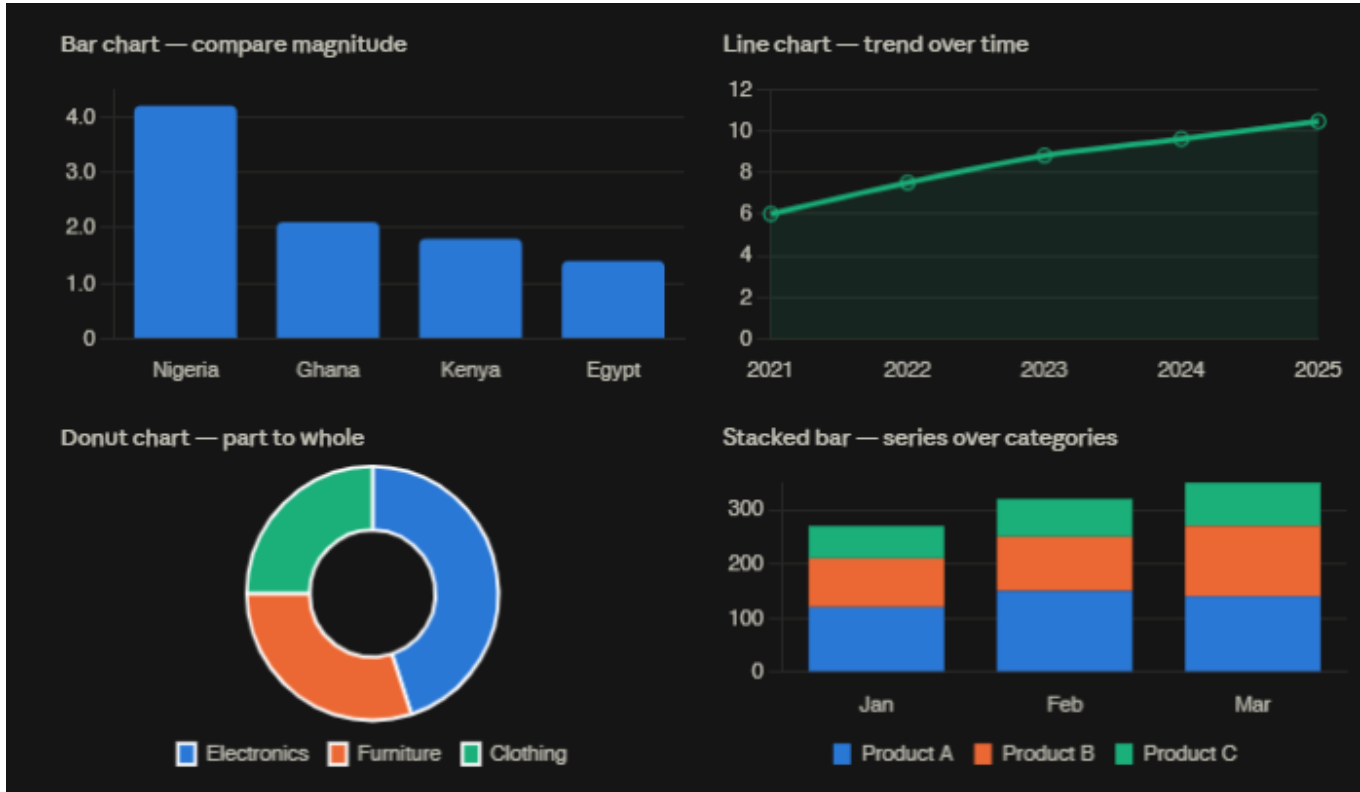
- Show totals
- Big numbers
- Quick summary

Charts

- Show breakdowns
- Explain the numbers
- Show comparisons

Example:

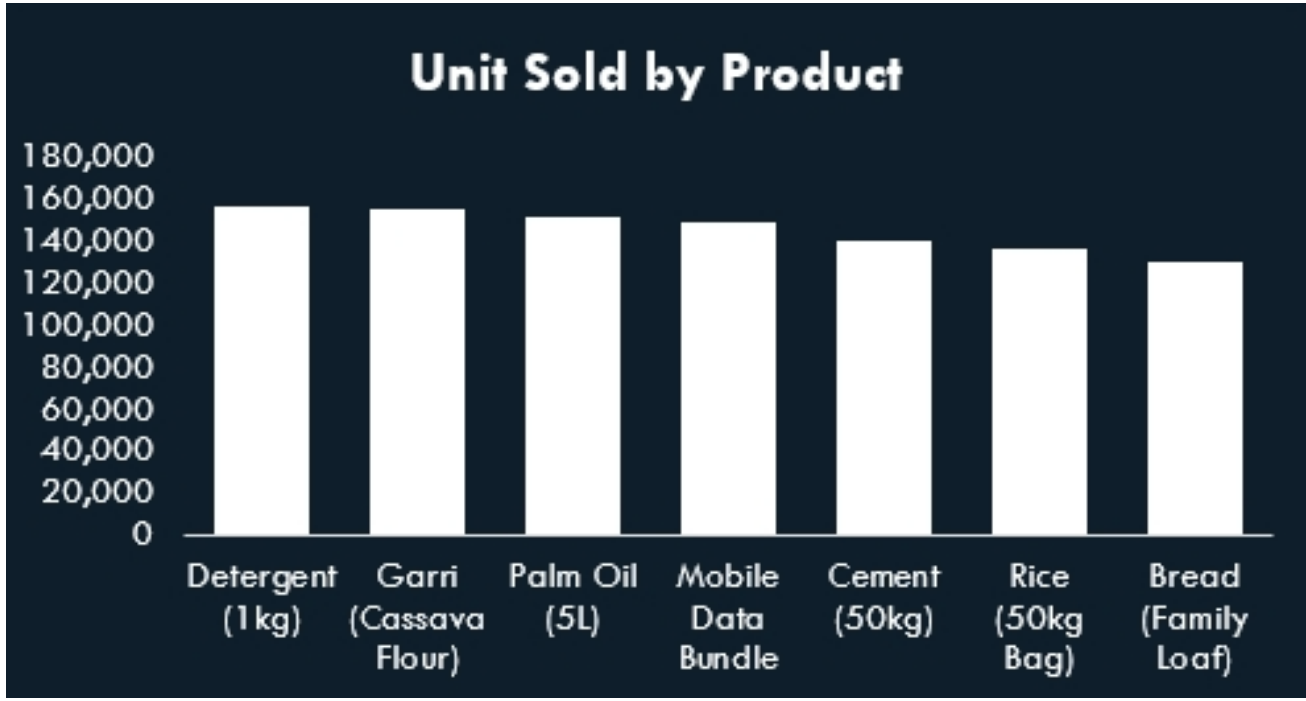
- KPI: Total Revenue
- Chart: Revenue by Country



One Chart, One Question
Each chart should answer one business question.

Example:

- ***Question***: Which country makes the most money?
- ***Chart***: Revenue by Country



COMMON DASHBOARD MISTAKES

Avoid:

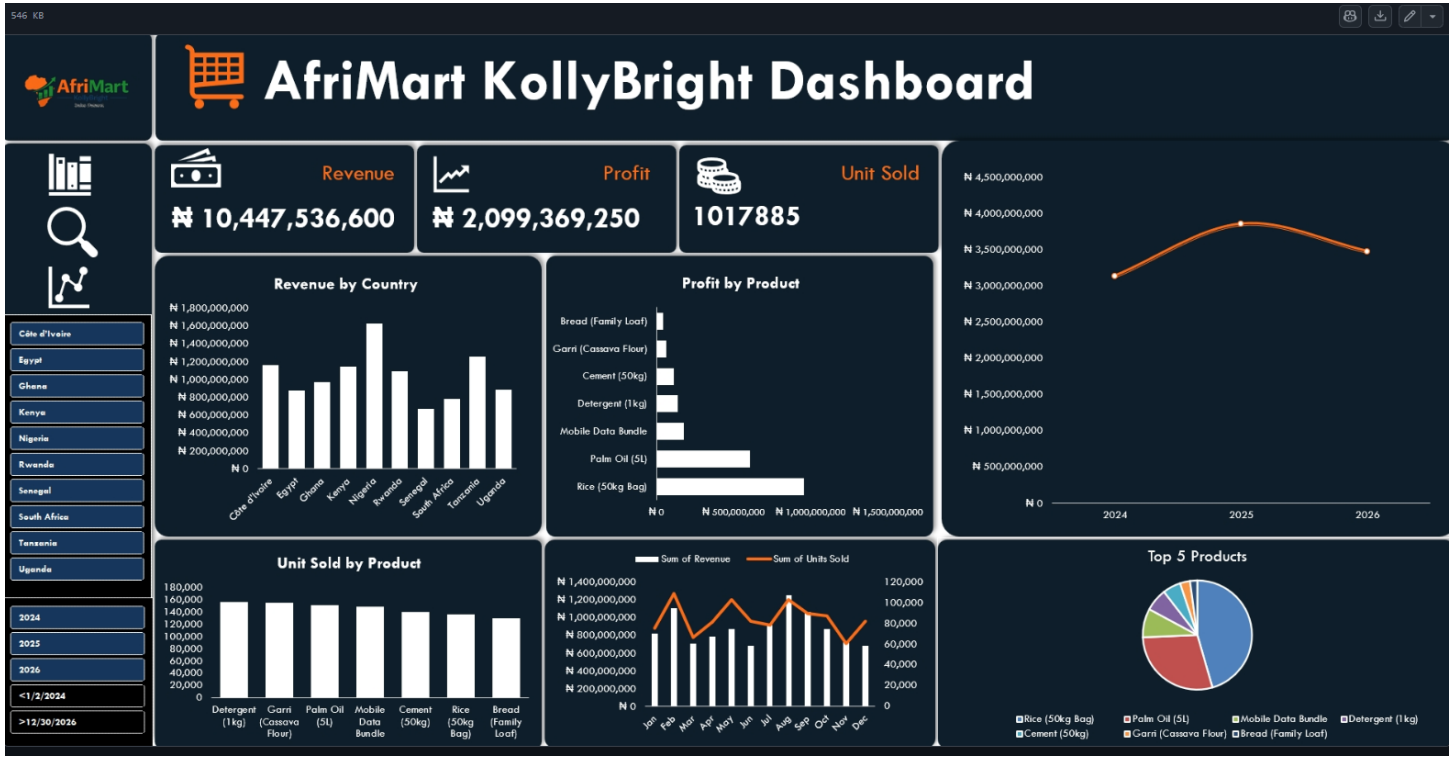
- Too many charts
- Too many colors
- No clear titles
- No KPIs

Keep dashboards clean and simple.

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Dashboard Layout (Best Practice)

1. KPIs at the top
2. Charts in the middle
3. Filters (slicers) on the side
4. Clear titles everywhere



Key Takeaway

A good dashboard:

- Is simple
- Answers business questions
- Uses KPIs, charts, and slicers correctly
- Can be understood in seconds

[Metric] by [Dimension(s)]

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Metric → the **number** being calculated (Revenue, Profit, Units Sold)

Dimension → how the number is broken down (Country, Product, Date)

DATA USED	CORRECT NAME
Sales + Country	**Sales by Country**
Revenue + Product	**Revenue by Product**
Profit + Product	**Profit by Product**
Units Sold + Country	**Units Sold by Country**

"We Always Start With The Number, Not The Category."

AfriMart sales dataset — before we start

The scenario

AfriMart is a retail and distribution company operating across 10 African countries. **Each row in the dataset is one sales transaction** — a specific product, sold in a specific country, on a specific date, with the resulting money numbers. This is called transaction-level data: nothing is pre-summarized. Every chart or KPI you build is just these rows added up in different ways.

What's being sold

7 everyday retail goods — Rice, Palm Oil, Detergent, Bread, Garri, Cement, and Mobile Data Bundles — spanning January 2024 to December 2026 (3 years), across Nigeria, Egypt, Kenya, Ghana, South Africa, Uganda, Rwanda, Senegal, Tanzania, and Côte d'Ivoire.

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Dataset structure (700 rows, 7 columns)

Column	Type	Role	What it means
Country	**Text**	**Dimension**	**Which of the 10 countries the sale happened in**
Product	**Text**	**Dimension**	**Which of the 7 products was sold**
Date	**Date**	**Dimension**	**When the sale happened (Jan 2024 – Dec 2026)**
Units Sold	**Number**	**Metric**	**Quantity of product sold in that transaction**
Revenue	**Number**	**Metric**	**Money earned from the sale**
Cost	**Number**	**Metric**	**Cost of acquiring/producing the goods sold**
Profit	**Number**	**Metric**	**Revenue minus Cost**

The one formula that explains every number

Revenue − Cost = Profit

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- Making the Dashboard dynamic
- Dashboard Structure
- Creating the charts and visuals
- Customising the visuals and putting it all together

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- ✓ **Left sidebar** → Navigation
- ✓ **Top bar** → Title / filters
- ✓ **Main content** → Charts, KPIs
- ✓ **Footer** → Notes / status

Dashboard Questions / Insights (Up to 8)

These are **realistic business questions** your dashboard can answer. Each can be visualized with a pivot table + chart:

Revenue by Country

Question: Which countries generate the most revenue?
Pivot: Rows = Country, Values = Revenue (Sum)
Chart: Column chart

Profit by Product

Question: Which products are most profitable?
Pivot: Rows = Product, Values = Profit (Sum)
Chart: Bar chart

Units Sold by Product

Question: Which products sell the most units?
Pivot: Rows = Product, Values = Units Sold (Sum)
Chart: Column chart

Revenue Trend Over Time

Question: How has revenue changed month to month or year to year?
Pivot: Rows = Date (group by Month/Year), Values = Revenue (Sum)
Chart: Line chart

Profit by Country

Question: Which countries are most profitable?
Pivot: Rows = Country, Values = Profit (Sum)
Chart: Column chart

Revenue by Country and Product

Question: Which products sell best in which countries?
Pivot: Rows = Country, Columns = Product, Values = Revenue (Sum)
Chart: Stacked column chart

Monthly Sales Performance

Question: Are there seasonal trends or peak sales months?
Pivot: Rows = Date (group by Month), Values = Revenue / Units Sold
Chart: Line or area chart

Top 5 Products by Profit

Question: Which 5 products generate the most profit?
Pivot: Rows = Product, Values = Profit, Sort descending, Top 5 filter
Chart: Bar chart

Top Section: KPIs
- Total Revenue
- Total Profit
- Total Units Sold
- Profit Margin (%)

Middle Section: Charts answering questions
- Revenue by Country
- Profit by Product
- Units Sold by Product
- Revenue Trend Over Time

Bottom / Side Section: Optional deeper analysis
- Revenue by Country and Product
- Top 5 Products by Profit
- Monthly Sales Performance

Slicers: Country, Product, Year

Project Builders
Build Real Experience

Before creating any dashboard, ask:

1.Who is the audience?
Executives, marketing, operations? Different users care about different metrics.
2.What questions need answering?
Only create visuals that answer business questions.
3.Which KPIs are critical?
Don't overload — 3–5 key metrics is enough.
4.How will interactivity help?
Use slicers for Country, Product, Year, etc.

KPIs (Up to 4)

KPIs are **big numbers at the top of your dashboard** that quickly tell the audience how the company is doing.
Based on your dataset (`Country`, `Product`, `Units Sold`, `Revenue`, `Cost`, `Profit`, `Date`), here are 4 KPIs you can use:

1.Total Revenue → Sum of Revenue across all countries and products
"Shows the total sales value for the business."
2.Total Profit → Sum of Profit across all countries and products
"Shows the money actually made after costs."
3.Total Units Sold → Sum of Units Sold
"Shows total volume of products sold."
4.Profit Margin (%) → Total Profit ÷ Total Revenue
"Shows how profitable the business is overall."



PART 2

Creating our Dashboard